

# 18: Tips for Case Studies

Stat 230 - Spring 2026

```
library(tidyverse)
library(ggformula)
library(ggResidpanel)
library(bayesrules)
library(patchwork)
```

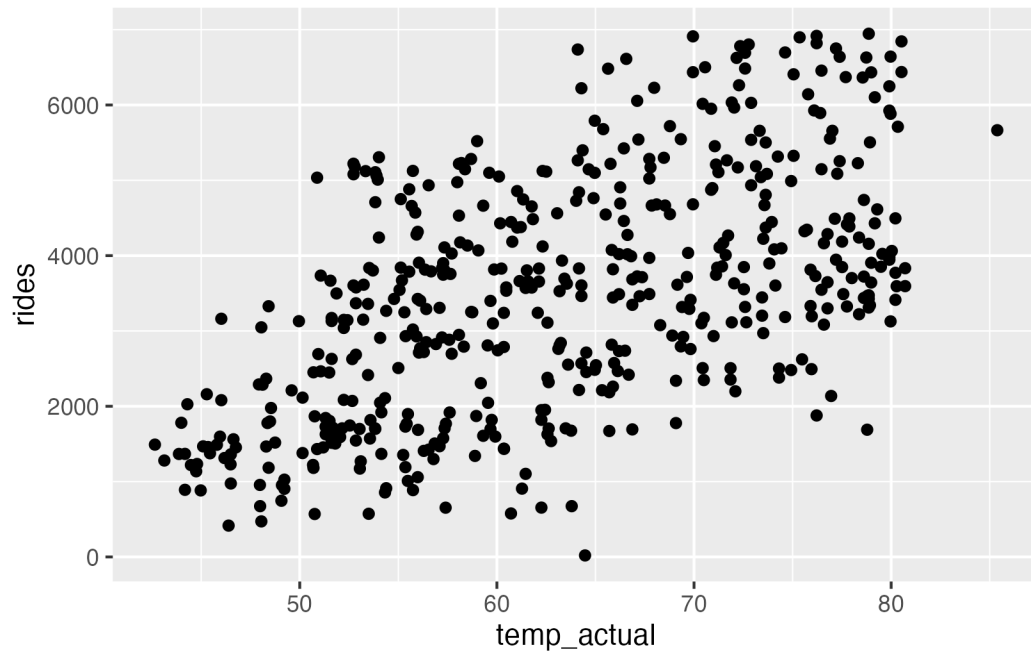
```
mod <- lm(rides ~ temp_actual, data = bikes)
```

## 1 Figure size

In the first few lines of your code chunks, you can specify figure output.

Default:

```
gf_point(rides ~ temp_actual, data = bikes)
```

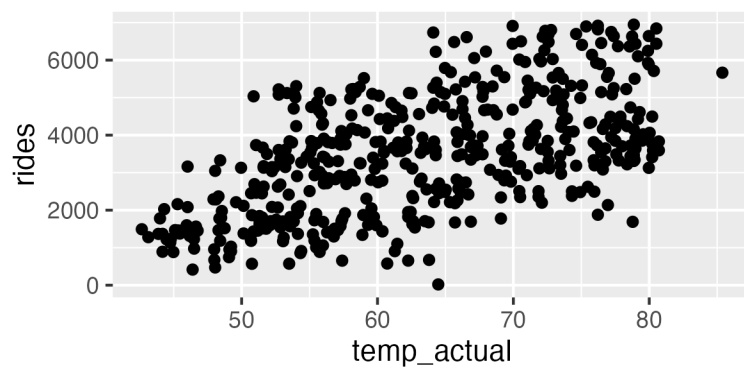


with

```
#l fig-width: 4  
#l fig-height: 2
```

at top of chunk:

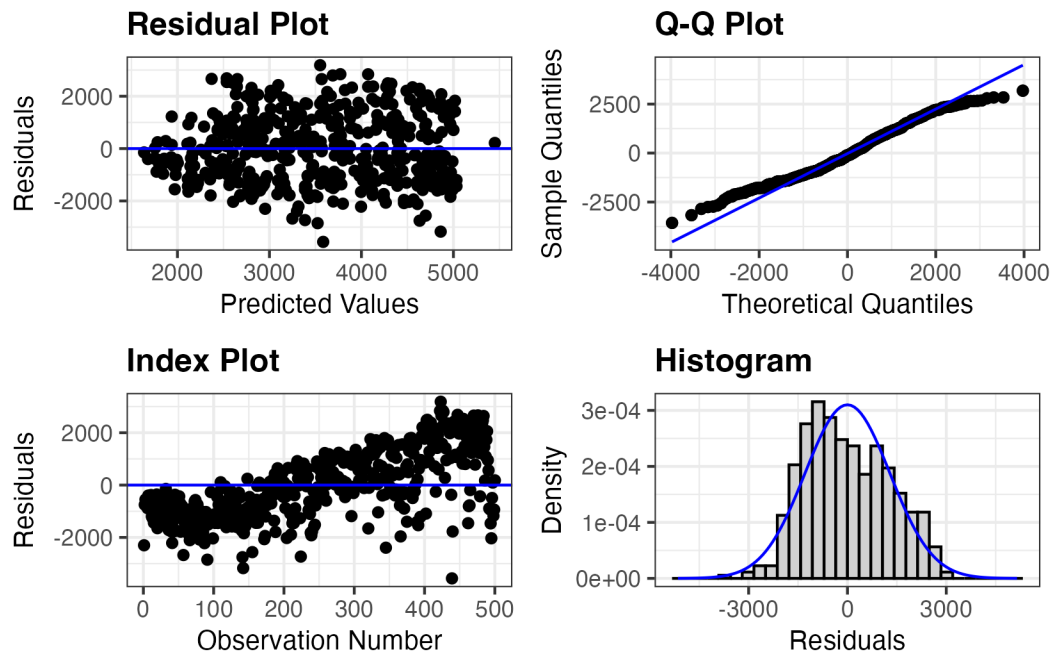
```
gf_point(rides ~ temp_actual, data = bikes)
```



## 2 Choosing a couple of residual plots

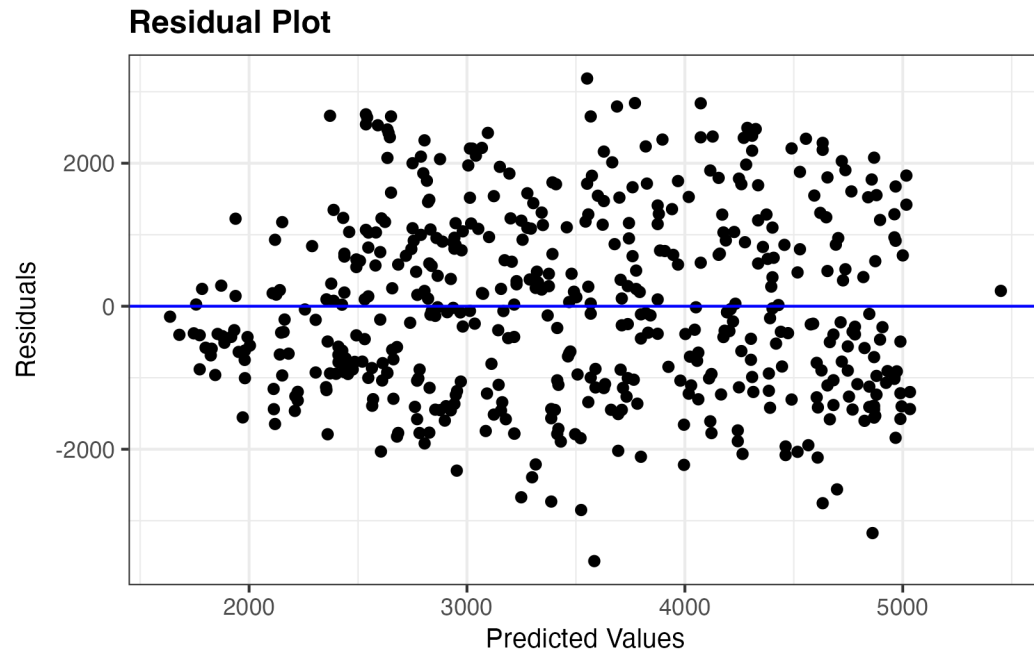
Sometimes, you might want to include some residual plots instead of the full panel (For example, you want to mention a weird trend in the index plot but all other residual plots are uninteresting). Specify using `plots` argument.

```
resid_panel(mod)
```



vs

```
resid_panel(mod, plots = "resid")
```

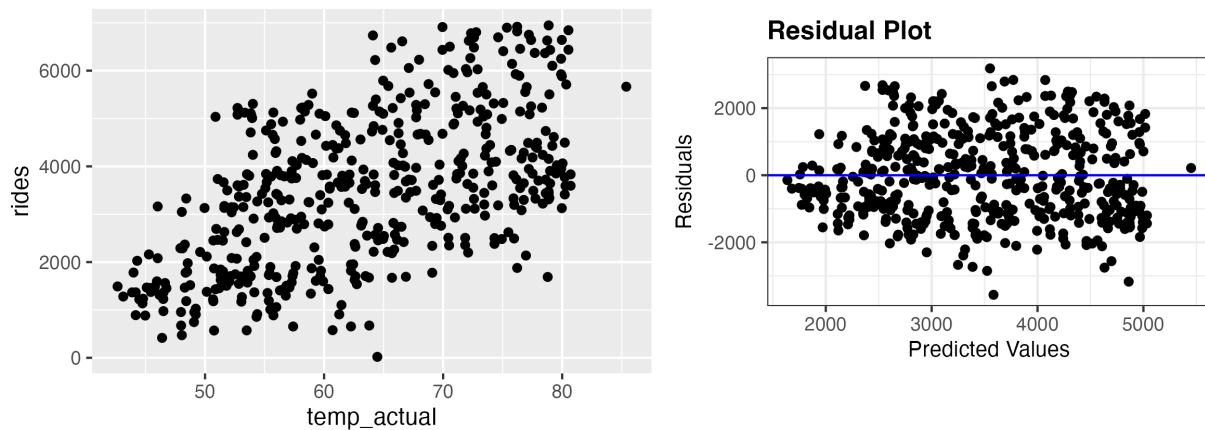


### 3 Combining graphics

Use the `{patchwork}` package to save individual plots and put them side by side

```
p1 <- gf_point(rides ~ temp_actual, data = bikes)
p2 <- resid_panel(mod, plots = "resid")

p1 + p2
```

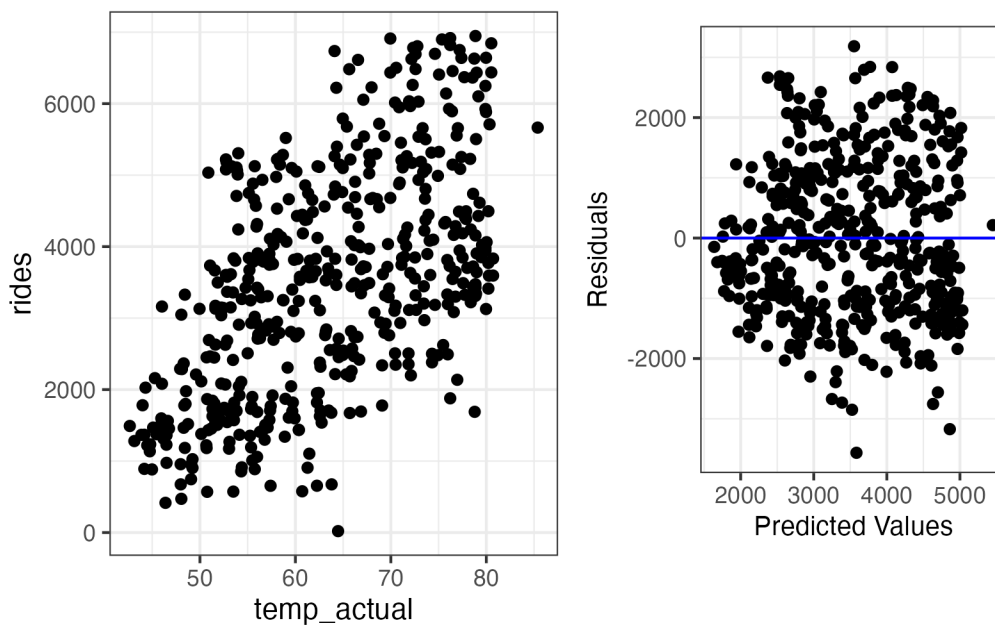


## 4 Customize theme

Oh no! The plots above look terrible together because they use different themes. *Note:* setting a theme is different with “regular plots” compared to resid panels

```
p1 <- gf_point(rides ~ temp_actual, data = bikes) + theme_bw()
p2 <- resid_panel(mod, plots = "resid", theme = "bw", title.opt = FALSE)

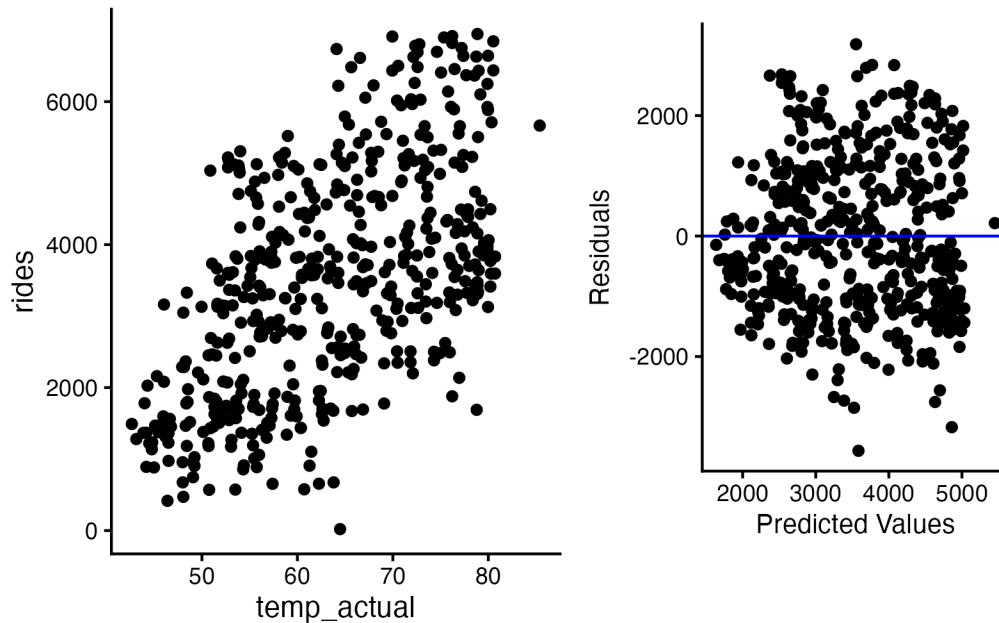
p1 + p2
```



Classic theme:

```
p1 <- gf_point(rides ~ temp_actual, data = bikes) + theme_classic()
p2 <- resid_panel(mod, plots = "resid", theme = "classic", title.opt = FALSE)

p1 + p2
```



*Note:* I know they show up as different sizes. I am not aware of a workaround besides using `augment()` and making your own residual plots.

## 5 Tables

The default summary output takes up too much space in a case study report:

```
summary(mod)
```

Call:

```
lm(formula = rides ~ temp_actual, data = bikes)
```

Residuals:

| Min     | 1Q      | Median | 3Q    | Max    |
|---------|---------|--------|-------|--------|
| -3564.3 | -1009.3 | -116.4 | 959.1 | 3184.6 |

Coefficients:

|             | Estimate | Std. Error | t value | Pr(> t )     |
|-------------|----------|------------|---------|--------------|
| (Intercept) | -2168.77 | 363.64     | -5.964  | 4.67e-09 *** |
| temp_actual | 89.23    | 5.67       | 15.739  | < 2e-16 ***  |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1288 on 498 degrees of freedom

Multiple R-squared: 0.3322, Adjusted R-squared: 0.3308

F-statistic: 247.7 on 1 and 498 DF, p-value: < 2.2e-16

Instead, use `broom::tidy()` for a more compact output.

```
broom::tidy(mod)
```

```
# A tibble: 2 x 5
```

|   | term        | estimate | std.error | statistic | p.value  |
|---|-------------|----------|-----------|-----------|----------|
|   | <chr>       | <dbl>    | <dbl>     | <dbl>     | <dbl>    |
| 1 | (Intercept) | -2169.   | 364.      | -5.96     | 4.67e- 9 |
| 2 | temp_actual | 89.2     | 5.67      | 15.7      | 1.35e-45 |

You can also specify confidence intervals:

```
broom::tidy(mod, conf.int = TRUE)
```

```
# A tibble: 2 x 7
```

|   | term        | estimate | std.error | statistic | p.value  | conf.low | conf.high |
|---|-------------|----------|-----------|-----------|----------|----------|-----------|
|   | <chr>       | <dbl>    | <dbl>     | <dbl>     | <dbl>    | <dbl>    | <dbl>     |
| 1 | (Intercept) | -2169.   | 364.      | -5.96     | 4.67e- 9 | -2883.   | -1454.    |
| 2 | temp_actual | 89.2     | 5.67      | 15.7      | 1.35e-45 | 78.1     | 100.      |

And to make it look EVEN nicer, pass it into `knitr::kable()`:

```
broom::tidy(mod, conf.int = TRUE) |>  
knitr::kable()
```

| term        | estimate    | std.error  | statistic | p.value | conf.low   | conf.high  |
|-------------|-------------|------------|-----------|---------|------------|------------|
| (Intercept) | -2168.77362 | 363.641906 | -5.964036 | 0       | -2883.2350 | -1454.3122 |
| temp_actual | 89.23389    | 5.669703   | 15.738723 | 0       | 78.0944    | 100.3734   |

which has a `digits` option:

```
broom::tidy(mod, conf.int = TRUE) |>
  knitr::kable(digits = 2)
```

| term        | estimate | std.error | statistic | p.value | conf.low | conf.high |
|-------------|----------|-----------|-----------|---------|----------|-----------|
| (Intercept) | -2168.77 | 363.64    | -5.96     | 0       | -2883.24 | -1454.31  |
| temp_actual | 89.23    | 5.67      | 15.74     | 0       | 78.09    | 100.37    |

## 6 Equations

If you are doing your report in .qmd, you might want to include some math! To include an inline equation, use single  $\$: y = \beta_0 + \beta_1 x_1 + \varepsilon, \varepsilon \sim N(0, \sigma^2)$ .

To give an equation its own line, use double  $\$$

$$y = \beta_0 + \beta_1 x_1 + \varepsilon, \varepsilon \sim N(0, \sigma^2)$$

(this typesetting language is called LaTeX and there are a bunch of resources available online)

- <https://latexeditor.lagrida.com/>
- <https://webdemo.myscript.com/views/math/index.html#>